

NATIONAL OCCUPATIONAL STANDARD FOR ARCHITECT



NOS.A.01
FIRST EDITION

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 20th December, 2023.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Construction NOSDT, upon which the following organisations were represented:

1. Copperbelt University
2. Engineering Institution of Zambia
3. Ministry of Local Government and Rural Development
4. Road Development Agency
5. Surveyors Institute of Zambia
6. University of Zambia
7. Zambia Institute of Architects
8. Association of Building and Civil Engineering Contractors
9. Water Resources Management Authority
10. Bari Zambia Limited
11. Zulu Barrow Construction
12. Ng'andu Consulting
13. Zambia Qualifications Authority (ZAQA) – Secretariat

ACKNOWLEDGEMENT

The Zambia Qualifications Authority would like to acknowledge the invaluable support of the following stakeholders that participated in the development of this NOS:

1. Arch. Mofya Kumisuku (Zambia Institute of Architects)
2. Arch. Peter Simphila Chiuto (Zambia Institute of Architects)
3. Dr. Erastus Mishengú Mwanaumo (University of Zambia)
4. Dr. Ephraim Zulu (Copperbelt University)
5. Mr. Abraham Zulu (Association of Building and Civil Engineering Contractors)
6. Mr. Felisian Ngosa (Ministry of Local Government and Rural Development)
7. Eng. Christopher Ngwira (Road Development Agency)
8. Dr. Chabota Kaliba (Engineering Institution of Zambia)
9. Mr. Michael Chileshe (Surveyors Institute of Zambia)
10. Mr. Stubbs Bernadine Mofya (Surveyors Institute of Zambia)
11. Mr. Douglas Lubaba (Water Resources Management Authority)
12. Eng. Dr. Kasongo Mwale (Ng'andu Consulting)
13. Mr. Elliot B. Phiri (Bari Zambia Limited)
14. Mr. Joseph Nyirenda (Zulu Barrow Construction)
15. Mr. Fidelis Cheelo (Zambia Qualifications Authority)
16. Mr. Jericho Kashiya (Zambia Qualifications Authority)
17. Ms. Womba Soneka (Zambia Qualifications Authority)

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Construction National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Construction sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Architecture is one of the critical trades in the construction industry. An Architect acts as a Designer, Project Co-ordinator, Project Manager, and Principal Agent for the client when undertaking his/her role on building or infrastructure project. He/she translates the client's requirements into architectural designs or information which he/she expresses through drawings or other documents based on which construction work is being undertaken—ranging from erecting of new buildings or infrastructure, to alterations, additions, rehabilitations, refurbishments, conversions, etc., of existing buildings or infrastructure. He/she often works with other related consultants such as Planners, Engineers, Quantity Surveyors, Geomatic Surveyors, and Environmentalist.

The Architect is trained to design buildings and infrastructure that correspond to the client's aesthetical, functional, cost, and time requirements that also adequately respond to the general environmental, social, cultural, economic factors, etc. The Architect acts as a key representative of the client during the construction process. He/she also provides post-construction services to clients relating to maintenance and management.

The development of this National Occupational Standard will ensure relevance of the training to latest advancements in industry, resulting in adequately and appropriately skilled Architects. This National Occupational Standard highlights the core knowledge, skills, competences and personal attributes that Architects must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

A	Architect
CS	Core Skills
K	Knowledge and Understanding
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.A.01
Occupation	Architecture
Job Title	Architect
Job Description	An Architect is responsible for the design of physical spaces and the management of the construction of such spaces.
Job Purpose	Architects respond to the clients' requirements for specific function spaces, translating their expressed needs into built environment fabric that is fit for purpose, and responds to environmental requirements.
ZQF Level	7
Sector	Construction
Sub sector	Real Estate and Infrastructure Construction
Other Economic Sector(s) in which the Occupation is Practiced	Mining, Manufacturing, Telecommunication, Energy, Education/Training, Agriculture, Health.
Other Similar Jobs that can be Performed in the Occupation	Facilities Management, Property Management, Property Development Appraisal/Advisory Services, etc.
Minimum Educational Job Entry Qualification(s)	Bachelor's Degree in Architecture or equivalent
Practicing License Requirements (if any)	Zambia Institute of Architects Corporate Membership
Training/RPL (Suggested)	Degree
Minimum Job Entry Age	21 Years
Prior Experience (Recommended)	2 Years Post-graduate experience interning under supervision of a Registered Architect
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Architects are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

The job requires an individual to have the ability to apply advanced mathematical, scientific, cultural, historical, geographical, artistic, and statistical principles to provide solutions. The job requires problem-solving skills and reasoning ability, capacity to communicate effectively and clearly, and must be self-motivated and a great team leader/worker. He/she must be creative, have ability to plan and prioritise, be quality conscious, be occupational health and safety orientated, be physically fit and courteous, and as well as have dexterity and well-developed mobility skills.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 5 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This Unit covers the skills and knowledge required by an Architect to develop briefs from a client's requirements and translate them into architectural drawings or documents]

Unit No.	01
Unit Title	Develop Briefs from Client's Requirements and Translate Into Architectural Drawings or Documents
Description	This Unit describes the skills and knowledge required to understand a client's requirements and translate them into spatial design sketches/drawings and specifications for constructional use.
Scope	This Unit covers the following: • Taking a Client's Instructions and Understanding the Requirements. • Translating the Client's Requirements into an Architectural Project Brief. • Translating the Project Brief into Architectural Drawings, Specifications or other architectural information for construction.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Taking a Client's Instructions and Understanding the Requirements	To be competent, the individual must be able to: PC1. Receive verbal or written instructions or requirements from a client. To do this, the individual needs to be able to: - Curate from the client's project requirements in writing or verbally; - Listen attentively, record accurately, read well, and comprehend information provided by the client; - Confirm with the client the instructions for accuracy and completeness.
Translating the Client's Requirements into an Architectural Project Brief	To be competent, the individual must be able to: PC2. Convert the client's requirements into an architectural project brief, and translate them into architectural design parameters.
Translating the Project Briefs into Architectural Drawings	To be competent, the individual must be able to: PC3. Convert the architectural design parameters for the production of architectural drawings, specifications, and/or other architectural information required for the project.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisation's policies, procedures, and operating manuals. OK2. The Zambia Institute of Architects Handbook.

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Design concepts/approach. - TK2. Construction materials performance and workmanship standards. - TK3. Local environmental conditions. - TK4. Planning legislation and regulations. - TK5. Relevant legislation guiding and providing restrictions with respect to architectural services and works. - TK6. Construction codes and mandatory standards. - TK7. Other consulting disciplines and services relevant to the project. - TK8. Available construction skills. - TK9. Project appraisal/feasibility. - TK10. Project brief preparation. - TK11. Preparation of architectural designs, specifications, and reports.
C. Regulatory Context (Practice License and Knowledge)	The individual on the job must demonstrate knowledge and understanding of the: - RK1. Zambia Institute of Architects Act and subsidiary legislation. - RK2. Urban planning legislation. - RK3. Public Health Act and subsidiary legislation. - RK4. Fire services regulations. - RK5. Lands Act and subsidiary legislation. - RK6. Environmental Management Act and subsidiary legislation. - RK7. National conservation and heritage legislation. - RK8. Occupational Health and Safety Act. - RK9. National Construction Council Act and subsidiary legislation. - RK10. Legislation and regulations governing other disciplines relevant to the project or assignment.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: - CS1. Write in English, and be able to translate the client's instructions into project briefs that can easily be understood by other staff working in the architectural office, as well as other project consultants providing input on the project. - CS2. Prepare and provide clear and comprehensive architectural drawings, details, and specifications that can be used for pricing and construction of the project.
	Reading Skills
	The individual on the job must be able to: CS3. Read in English, comprehend the client's instructions, and develop project briefs that can easily be understood by staff working in the drawing office or architectural firm.

	CS4. Read and interpret sketches, drawings, photographs, illustrations, etc., provided by the client that provide illumination to their requirements. CS5. Read and interpret various planning and building regulations, and other user specific construction guidelines, etc., of relevance to the project.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS6. Speak in English, and be able to give instructions in the office. CS7. Speak in English, and be able to give instructions in the local language used at the site. CS8. Listen attentively and interpret communication/ instructions from the client to co-workers, consultant's team, and contractors.
B. Professional Skills	Decision Making The individual on the job must be able to: PS1. Interpret the client's instructions or requirements into spatial designs that are fit for purpose.
	Plan and Organise The individual on the job should be able to: PS2. Plan work and organise the preparation of all required project documentation, approvals/permits, in coordination with team members, sub-consultants, and stakeholders.
	Customer Centricity The individual on the job must be able to: PS3. Accurately curate and translate the client's requirements into architectural drawings and information that can be used to construct a building or infrastructure that meets their requirements and vision. PS4. Observe the highest level of ethical conduct and confidentiality.
	Problem-Solving The individual on the job should be able to: PS5. Resolve any technical and administrative conflicts within the team and with external stakeholders.
	Analytical Thinking The individual on the job should be able to: PS6. Analyse and determine how the client's requirements can be best translated into spatial designs and architectural construction information.
	Critical Thinking The individual on the job should be able to: PS7. Identify and avert any violation of safety and other special spatial design principles, planning/building regulations, building contract infringements, etc.

UNIT 2 [This Unit covers the skills and knowledge required by an Architect to provide project budget estimates with respect to time and cost]

Unit No.	02
Unit Title	Project Budget Estimates Related to Time and Cost
Description	This Unit describes the skills and knowledge required to understand a client's requirements and translate them into financial cost estimates and estimated project durations.
Scope	This Unit covers the following: • Providing Cost Estimates (based on available data gathered from similar projects, current or historical, considered in combination with prevailing market and economic factors). • Providing Estimated Project Durations (based on available data with respect to similar projects executed, available construction technology, materials, logistical requirements, new industry innovations, etc.).
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Providing Cost Estimates	To be competent, the individual must be able to: PC1. Prepare project cost estimates using works cost rates obtained from Quantity Surveyors, construction works pricing indices, and similar recent or current projects, and apply appropriate rates considered in combination with prevailing market and economic factors to arrive at realistic cost estimates.
Providing Estimated Project Durations	To be competent, the individual must be able to: PC2. Gather resources and skills information from the construction industry from current or historical projects, and apply appropriate assumptions in order to arrive at project duration estimates.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisation's policies, procedures, and operating manuals as they relate to project cost estimation and project duration. OK2. The Zambia Institute of Architects Handbook as it relates to project cost estimation and project duration.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Construction costs and rates. TK2. Relevant consultants' fees. TK3. Applicable statutory fees. TK4. Construction processes. TK5. Sequencing and task durations, etc.
C. Regulatory Context	The individual on the job must demonstrate knowledge and understanding of the:

	RK1. Zambia Institute of Architects Act and subsidiary legislation. RK2. Relevant legislation pertaining to project team consultants' fees.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Generate a spreadsheet providing high level cost estimates of key project components. CS2. Generate a project programme highlighting key milestones of the project's life cycle and durations.
	Reading Skills
	The individual on the job must be able to: CS3. Read in English, and be able to comprehend the client's instructions, and translating them into project components with related cost estimates. CS4. Read in English, and be able to comprehend the client's instructions translating these into project activities with estimated durations.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Speak in English, and present project cost estimates to the client and any other stakeholders. CS6. Speak in English, and present estimated project durations, activities or programmes to the client and any other stakeholders.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Interpret the client's instructions or requirements into spatial design solutions, fitting into their budget, delivery time, and vision.
	Plan and Organise
	The individual on the job should be able to: PS2. Plan and organise activities, workflows, and resources related to gathering information, in collaboration with other relevant consultants that would result in the development of accurate and realistic cost estimates and project duration estimates.
	Customer Centricity
	The individual on the job must be able to: PS3. Obtain rates for various work components, using current and historical projects as references in consultation with other industry experts, and apply prevailing market and/or economic conditions to arrive at realistic and accurate project budget estimates. PS4. Observe the highest level of ethical conduct and confidentiality.

	Problem-Solving
	The individual on the job should be able to: PS5. Recognise threats to the project budget and duration, and recommend measures to avert cost escalations and project delays, or should be able to recommend specific measures to de-escalate costs and make up for lost time.
	Analytical Thinking
	The individual on the job should be able to: PS6. Analyse and determine how the client's requirements can be translated into project components, activities, and processes with associated costs and timelines.
	Critical Thinking
	The individual on the job should be able to: PS7. Identify and avert any risks with respect to cost escalation and project completion delays.

UNIT 3 [This Unit covers the skills and knowledge required by an Architect for greenfield of design spaces, buildings or infrastructure, and conversion, refurbishment, alteration, restoration, rehabilitation of existing spaces, buildings or infrastructure]

Unit No.	03
Unit Title	Greenfield Architectural Design Services
Description	This Unit describes the skills and knowledge required by an Architect for greenfield architectural design of spaces, buildings or infrastructure, and conversion, refurbishment, alteration, restoration, rehabilitation of existing spaces, buildings or infrastructure.
Scope	This Unit covers the following: • Providing Architectural Services for Greenfield Projects. • Providing Architectural Services for Existing Buildings or Infrastructure.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Providing Architectural Services for Greenfield Projects	To be competent, the individual must be able to: PC1. Produce architectural drawings or architectural information based on the client's brief or requirements, and scope of engagements agreed with the client to do this. The individual needs to be able to: - Curate from the client's project requirements. - Produce a Project Brief from the client's requirements. - Undertake field and desktop surveys, and research with respect to site conditions. - Demonstrate understanding of the local environmental and regulatory conditions, and restrictions impacting the successful delivery of the project. - Provide high-level cost estimates of project costs across the design stages as more detailed information is produced or becomes available. - Produce architectural drawings with the appropriate level of detail or information at each design stage sufficient to assist the client, consulting team members, regulators, and any other stakeholders to make decisions, or to take action that supports the successful delivery of the project. - Advise the client and obtain their consent with respect to the participation of other required professionals necessary for the successful delivery of the project. - Organise and co-ordinate the production, communication, and exchange of information amongst project team professionals. - Advise the client on the life-cycle, maintenance requirements, and environmental impact of the materials and construction technology incorporated in the fabric of the building or infrastructure.

Providing Architectural Services for Existing Buildings or Infrastructure	To be competent, the individual must be able to: PC2. Produce architectural drawings or architectural information based on the client's brief or requirements, and the scope of engagement agreed with them to do this. The individual needs to be able to: - Curate from the client's project requirements. - Produce a Project Brief from the client's requirements. - Undertake field and desktop surveys, and research with respect to existing buildings or infrastructure and site conditions. - Have an understanding of the local heritage, environmental and regulatory conditions, and restrictions impacting the successful delivery of the project. - Plan, organise, and manage engagement with multiple stakeholders, such as local communities, heritage, environmental, traffic management authorities, etc. - Provide high-level cost estimates of the project's costs across the design stages as more detailed information is produced or becomes available. - Produce architectural drawings with the appropriate level of detail or information at each design stage sufficient to assist the client, consulting team members, regulators, and any other stakeholders to make decisions, or to take action that supports the successful delivery of the project. - Advise the client and obtain their consent with respect to the participation of other required professionals necessary for the successful delivery of the project. - Organise and co-ordinate the production, communication, and exchange of information amongst project team professionals. - Advise the client on the life-cycle, maintenance requirements, and environmental impact of the materials and construction technology incorporated in the fabric of the buildings or infrastructure.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisation's policies, procedures, and operating manuals as they relate to architectural design services. OK2. The organisation's design ethos and philosophy with respect to the architectural design approach and methodology. OK3. The Zambia Institute of Architects Handbook as it relates to architectural design services.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Spatial design theories, principles, and guidelines.

	TK2. Local environmental, cultural, heritage, organisational factors affecting the project, and the design and construction. TK3. Relevant legislation, regulations, and consents or restrictions affecting the project. TK4. Sustainable design solutions and sustainable construction methodologies. TK5. Construction technology, services, and skills. TK6. Construction materials and their performance in the local environment. TK7. Post-construction operations and maintenance requirements of the design solution, materials, construction methodology, and the likely user conduct or behaviours. TK8. Cost impact of the design solutions or material choices (working knowledge). TK9. Project programming, sequencing, and durations. TK10. Trends, styles, culture, etc., that may impact the performance of the architectural solution provided. TK11. CAD packages. TK12. Digital field instruments, e.g., Laser Range Finders, Laser levels, Drones, etc.
C. Regulatory Context	The individual on the job must demonstrate knowledge and understanding of the: RK1. Zambia Institute of Architects Act and subsidiary legislation. RK2. Urban planning legislation. RK3. Public health legislation. RK4. Fire safety and services legislation. RK5. Lands legislation. RK6. Environmental management legislation. RK7. National conservation and heritage legislation. RK8. Occupational Health and Safety legislation. RK9. Construction legislation. RK10. Any other legislation relevant to the project.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Generate an accurate and informative description of the materials to be used and installation methodologies. CS2. Generate accurate and appropriate specifications for the materials, workmanship, and reference building codes or standards for work's execution.
	Reading Skills
	The individual on the job must be able to: CS3. Read in English, and be able to comprehend the materials' technical data information.

	CS4. Read in English, and be able to comprehend the workmanship and reference building codes or standards for the work's execution.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to:
	CS5. Speak in English, and present design information to the client, contractors, relevant stakeholders, and site staff in at least one local language.
B. Professional Skills	Decision-Making
	The individual on the job must be able to:
	PS1. Interpret the client's instructions or requirements into spatial design solutions fitting into their requirements and vision.
	Plan and Organise
	The individual on the job should be able to:
	PS2. Plan and organise design processes, workflows, durations, and required resources (financial, technological, human, etc) for the assignment in accordance with the available budget and allocated time.
	Customer Centricity
	The individual on the job must be able to:
	PS3. Translate the client's requirements into architectural design solutions that answer to their spatial needs, vision, budget, and delivery times.
	PS4. Observe the highest level of ethical conduct and confidentiality.
	Problem-Solving
	The individual on the job should be able to:
	PS5. Resolve design challenges innovatively within the agreed project budget and duration.
	Analytical Thinking
	The individual on the job should be able to:
	PS6. Analyse and determine how the client's requirements can be translated into appropriate spatial design solutions that are congruent with their vision.
	Critical Thinking
	The individual on the job should be able to:
	PS7. Identify design solutions that align with the client's spatial requirements, budget, and delivery timeline.

UNIT 4 [This Unit covers the skills and knowledge required by an Architect to provide quality control, quality assurance, project monitoring, project management, and team leadership]

Unit No.	04
Unit Title	Supervision/Project Management Services – Quality Control and Project Monitoring
Description	The Unit describes the skills and knowledge required to provide quality control, quality assurance, project monitoring, project management, and team leadership.
Scope	This Unit covers the following: • Providing Project Team Leadership. • Providing Quality Control Systems. • Providing and Putting in Place Quality Assurance Measures. • Providing Project Monitoring Services (including tracking of progress). • Providing Contract Management Services, such as Guidance on Contractual Obligations, Dispute Resolution Services (arbitration, expert witness services). • Certification of Payments. • Certification of Performance Stages, such as Site Handover, PC, Final Account. • Liaison with Regulators Impacting the Projects for Acquisition of Permits, Stage Inspections, and other necessary consents/approvals.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Providing Project Team Leadership	To be competent, the individual must be able to: PC1. Demonstrate understanding of other disciplines and the resources required to deliver solution to the client's requirements. PC2. Demonstrate communication and interpersonal skills. PC3. Be ethical and impartial. PC4. Exhibit knowledge of sequencing of activities, milestones, and deliverables. PC5. Understand the various project consultants, their roles, responsibilities, and contractual obligations. PC6. Comprehend relevant legislation and regulations likely to impact the solutions. PC7. Apply timely decision-making, communication, and implementation of agreed actions.
Providing Quality Control systems	To be competent, the individual must be able to: PC8. Understand all the deliverables of the project consultants in order to avoid gaps or clashes, and ensure comprehensive quality control on the project. PC9. Exhibit working knowledge of the types of inspections, tests, observations, outputs, and reports that each project team member should produce.

	PC10. Ensure ethics and impartiality in the determination of matters that may arise within the project team. PC11. Demonstrate knowledge of sequencing of events and actions required to deliver solutions. PC12. Comprehend relevant legislation and regulations likely to impact the solutions.
Providing and Putting in Place Quality Assurance Measures	To be competent, the individual must be able to: PC13. Understand and implement quality assurance management systems required on projects, output reports, remediation systems, and anticipated outcomes.
Providing Project Monitoring Services (including tracking of progress)	To be competent, the individual must be able to: PC14. Demonstrate working knowledge of documentation and processes accompanying the delivery of solutions. PC15. Measure, track, and record progress against baseline project monitoring information or tools.
Providing Contract Management Services, such as Guidance on Contractual Obligations, Dispute Resolution Services (arbitration, expert witness services)	To be competent, the individual must be able to: PC16. Comprehend various types of contracts and contract management/administration skills. PC17. Apply timely decision-making, communication, and implementation of agreed actions. PC18. Ensure impartiality with respect to interpretation of contractual clauses and terms when determining issues that may arise between contractors and the client, or with project team consultants, including the individual. PC19. Comprehend alternative dispute resolution mechanisms for the appropriate type of disputes.
Certification of Payments	To be competent, the individual must be able to: PC20. Exhibit knowledge of the benchmarks and milestones that trigger certification of valuations prepared by Quantity Surveyors in favour of contractors.
Certification of Performance Stages, such as Site Handover, PC, Final Account	To be competent, the individual must be able to: PC21. Understand the benchmarks and milestones that trigger stage inspections, certifications, such as stage inspections, payment certificates, practical completion, site handover, partial possession, project completion certificates, etc., and project final accounts.
Liaison with Regulators Impacting the Project for Acquisition of Permits, Stage Inspections, and other necessary consents/approvals	To be competent, the individual must be able to: PC22. Possess good working knowledge of the wide range of stakeholders, including neighbours, communities, regulators, government departments, agencies, and regulators. PC23. Possess good working knowledge of the rights, roles, mandates, legal status of stakeholders, and the recourse that they have if they object to the project. PC24. Demonstrate working knowledge of the processes and deliverables with respect to engagement of stakeholders, acquisition of consents, sign-offs, and certifications that impact the project.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisation's policies, procedures, and operating manuals as they relate to project supervision and project management services. OK2. The organisation's design ethos and philosophy in respect of quality control and quality assurance. OK3. The Zambia Institute of Architects Handbook as it relates to architectural design services.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Law of Contracts. TK2. Construction Law. TK3. Construction Contract Management. TK4. Project Management. TK5. Roles and responsibilities of various professionals or project team members that may be required to support the architectural solutions or deliver the project. TK6. Construction standards and codes. TK7. Architecture, Planning, Public Health, Environmental Management, Fire Safety, Safety and Health, Local Government, and any other relevant legislation and regulations. TK8. Various types of construction contracts. TK9. CAD software packages. TK10. Project programming software packages. TK11. Digital field instruments, e.g., Laser Range Finders, Laser levels, Drones, etc. TK12. Dispute resolution mechanisms. TK13. Quality Control and Quality Assurance.
C. Regulatory Context (Practice License and Knowledge)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Zambia Institute of Architects Act and subsidiary legislation. RK2. Urban planning legislation. RK3. Public Health Act and subsidiary legislation. RK4. Fire services regulations. RK5. Lands Act and subsidiary legislation. RK6. Environmental Management Act and subsidiary legislation. RK7. National conservation and heritage legislation. RK8. Occupational Health and Safety Act. RK9. National Construction Council Act and subsidiary legislation. RK10. Legislation and regulations governing other disciplines relevant to the project or assignment.

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Write in English, and generate informative and accurate correspondence and reports. CS2. Write in English, and generate project programmes, and track activities highlighting achievements, delays, key milestones, etc. of the project.
	Reading Skills The individual on the job must be able to: CS3. Read in English, and be able to comprehend client, contractor, consultant, regulator, etc. communications, and assess implications and impacts on the project.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS4. Speak in English, and provide guidance and information to project team members during project meetings, site meetings, and coordination discussions. CS5. Speak in English, and at least one local language for communication of instructions to site staff.
	Decision-Making The individual on the job must be able to: PS1. Interpret situations arising on projects, and provide guidance on how to facilitate progress, mitigate risks, resolve disputes, facilitate honouring of contractual obligations by the parties, etc., for the successful delivery of the project.
B. Professional Skills	Plan and Organise The individual on the job should be able to: PS2. Plan and organise project activities, workflows, durations and required resources (financial, technological, human, etc.) required for the assignments in accordance with agreed contract sums and contract periods. PS3. Plan and organise quality control and quality assurance activities, and put in place tools and resources for checking, recording, and certification of progress achieved or lack thereof. PS4. Plan and organise activities, check points to facilitate risk assessment, and put in place mitigative measures to facilitate the project's progress and adherence to agreed quality, time, and cost parameters.
	Customer Centricity The individual on the job must be able to: PS5. Focus all the actors and resources on the project toward delivery within the cost, time, quality, and vision agreed with the client.

	PS6. Galvanise the actors towards observing the highest levels of ethical conduct, ensuring that the client gets value for their money.
	Problem-Solving
	The individual on the job should be able to:
	PS7. Recognise threats or risks to the project, cost management, and contract durations, and recommend measures to avert cost escalations, project delays or disputes.
	PS8. The individual should be able to recommend specific measures to avert or manage any risks to the projects.
	Analytical Thinking
	The individual on the job should be able to:
	PS9. Identify and determine how the resources, information, and efforts can be focused toward the successful delivery of the project.
	Critical Thinking
	The individual on the job should be able to:
	PS10. Identify and avert any risks with respect to delivery of the project within agreed cost and time.

UNIT 5 [This Unit covers the skills and knowledge required by an Architect to provide various consultancy services requiring architectural expertise]

Unit No.	05
Unit Title	Various Architectural Consultancy Services
Description	This Unit covers the skills and knowledge required by an Architect to provide various consultancy services requiring the expertise of an Architect.
Scope	This Unit covers the following: • Architectural Valuations. • Sectional Titling. • Property Development Appraisals. • Property Maintenance Management Services. • Architectural Design Competitions. • Expert Witness Services. • Alternative Dispute Resolution, etc.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Architectural Valuations	To be competent, the individual must be able to: PC1. Survey and gather information on the state of buildings or infrastructure for the intended valuation purposes. PC2. Prepare reports detailing the required actions or works, and associated financial information, such as repairs or reinstatement of cost estimates, etc.
Sectional Titling	To be competent, the individual must be able to: PC3. Survey and gather information relating to the demising of the sections of the properties, identification of shared building elements, common areas, common accesses, and utility services, etc. PC4. Prepare reports with accompanying diagrams for use in the issuance of sectional certificates of title to the building's owner by the relevant authorities.
Property Development Appraisals	To be competent, the individual must be able to: PC5. Provide sound and accurate advice to prospective property developers, with respect to building and infrastructure design trends, styles, materials, construction techniques and methodologies; construction costs and project delivery timelines, etc. PC6. Advise developers on design and construction investment parameters and considerations, taking into account historical performances of similar developments, current circumstances, and possible future trends, developments and innovations.

Property Maintenance Management Services	To be competent, the individual must be able to: PC7. Survey and gather information on state of repairs of the property at project inception. PC8. Gather designs and as-built information relating to the property where available. PC9. Recommend remedial and maintenance measures to refurbish, restore, improve, and maintain the property to facilitate peak performance as per design intent.
Architectural Design Competitions	To be competent, the individual must be able to: PC10. Provide guidance to architectural competition promoters with respect to regulations governing such competitions and their adjudication.
Expert Witness Services	To be competent, the individual must be able to: PC11. Understand architectural designs, materials, construction technologies and methodologies, workmanship standards, building codes, regulations, and construction consultancy contracts, and construction contracts. PC12. Provide professional unbiased, accurate, and truthful testimonies in court or any other dispute resolution forums with respect to understanding of the matters, and how they may be resolved as per the agreements of the parties at inception, or as modified in the course of their relationships.
Alternative Dispute Resolution	To be competent, the individual must be able to: PC13. Understand architectural designs, materials, construction technologies and methodologies, workmanship standards, building codes, regulations, and construction consultancy contracts, and construction contracts. PC14. Provide professional unbiased, accurate, and truthful mediations or adjudications in dispute resolution forums with respect to disputes brought before them; understand matters and how they could be resolved as per the agreements of the parties at inception, or as modified in the course of their relationships.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisation's policies, procedures, and operating manuals as they relate to other architectural consultancy services.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Law of Contracts. TK2. Construction Law. TK3. Various types of construction contracts. TK4. Construction Management. TK5. Project Management.

	TK6. Roles and responsibilities of various professionals or project team members that may be required to support the architectural solution or deliver projects. TK7. Construction standards, codes, technology, materials, and methodologies. TK8. Architectural design theories, principles, and trends. TK9. Relevant legislation. TK10. Economics (working knowledge). TK11. Construction cost estimations.
C. Regulatory Context	The individual on the job must demonstrate knowledge and understanding of the: RK1. Zambia Institute of Architects Act and subsidiary legislation. RK2. Public Health Act and subsidiary legislation. RK3. Fire services regulations. RK4. Environmental Management Act and subsidiary legislation. RK5. National conservation and heritage legislation. RK6. Occupational Health and Safety Act. RK7. Lands Act. RK8. Any other relevant legislation.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and generate various types of technical reports.
	Reading Skills
	The individual on the job must be able to: CS2. Read in English, and be able to comprehend the client's instructions and other relevant information, translating these into required outputs.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS3. Speak in English, and present information, findings or reports to the client and any other stakeholders.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Consider the client's instructions or requirements, and the information gathered during the assignments, and provide them with professional and accurate feedback or outputs as required by the assignment parameters.
	Plan and Organise
	The individual on the job should be able to: PS2. Plan and organise project activities, workflows, durations, and required resources (financial, technological, human, etc) required for the assignments in accordance with available budgets and allocated project durations.

	Customer Centricity
	The individual on the job must be able to: PS3. To provide professional, accurate, and efficient services as pertains to the required consultancy assignments within the agreed costs and delivery times. PS4. To observe the highest levels of ethical conduct and confidentiality.
	Problem-Solving
	The individual on the job should be able to: PS5. Recognise threats to project budgets and durations, and recommend measures to avert cost escalations and delays, or should be able to recommend specific measures to de-escalate costs, and to make up for lost time.
	Analytical Thinking
	The individual on the job should be able to: PS6. Analyse and determine how the client's requirements can be translated into project deliverables, activities, and processes within associated costs and timelines.
	Critical Thinking
	The individual on the job should be able to: PS7. Identify and avert any risks with respect to providing erroneous or faulty services, cost escalations, and project completion delays.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

Equipment, tools, and consumable materials used by the jobholder include, but are not limited to:

- ICT
- CAD Packages
- Microsoft Office Packages
- MS Projects or equivalent
- Digital Measuring Field Equipment
- Photography Equipment
- Drones

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Dilemmas associated with the job of an Architect include:

- Working in dangerous areas – uncleared bush; at height, etc
- Exposure to construction hazards such as noise, heat, cold, chemicals, etc.;
- Working long hours;
- Frequent travel;
- Driving long distance on bad roads, including at night;
- Temptation to accept gratification from contractors, materials suppliers, litigants, etc., as inducement to make decisions in their favour, or for award of contracts or for works not done to specifications;
- Pressure from politicians to make unprofessional decisions or rulings that may be politically expedient;
- Pressure from clients, colleagues, supervisors, contractors, regulators, etc. to make decisions without fully considering outcomes or consequences in the interest of making progress on projects.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include: wearing protective clothing, and ensuring the availability and use by other employees; exercising regularly to maintain physical fitness; exercising proper work ergonomics; participating in workplace safety sensitisation and awareness meetings/training sessions; adhering to institutional safety and standard operating procedures at all times; consulting extensively within and outside one's department/ team on construction safety issues; planning and prioritising work; professional integrity and ethics; stakeholder engagement, people management training, and consultations, etc.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include: indoor and outdoor construction sites, as well as workshops and factories; may also work in commercial buildings or private homes, and confined spaces; handling machines with moving parts; working at great heights; working in conditions that may be dirty and noisy; exposure to seasonal

heat and cold, or adverse weather conditions; emergency call-outs; standing or squatting for long hours; lifting relatively heavy objects; In most cases, the job involves working normal hours, but in some instances, long working hours and/or regular overtime may be required. The job also requires wearing suitable protective clothing, such as works suits, safety shoes, ear protectors, safety visors or goggles, gloves, and hard-hats, safety harnesses, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Parties involved or interacting with the jobholder that are internal to the organisation, and may include: firm owners, supervisors/superiors, trainers, occupational health and safety team, other colleagues, etc.

8.2 External/Outside the Organisation

Parties involved or interacting with the jobholder that are external to the organisation, and may include: government regulators, trainers, clients, suppliers of equipment/tools/consumables, fellow Architects from other companies, Engineers, Planners, Quantity Surveyors, Contractors, materials manufacturers, equipment suppliers, labour unions, occupational health and safety agencies, stakeholder groups, etc.

9. PHYSICAL DEMANDS ON THE BODY

Ability to sustain strenuous conditions such as:

- Climbing heights;
- Walking and standing for long periods of time;
- Bending, stretching, twisting, or reaching out;
- Lifting, carrying, pushing and pulling heavy objects;
- Using fingers, hands, and feet with ease to complete the assigned tasks (dexterity);
- Strenuous works that may cause musculoskeletal disorders;
- Working in front of computer screens may cause eyesight defects;
- Etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for the skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.A.01		
ZQF Level	7	Version Number	01
Sector	Construction	Date of Approval	December 2023
Sub Sector	Real Estate and Infrastructure Construction	Date of Last Review	N/A
Occupation	Architect	Date of Next Review	December 2028



*“You learn,
We Standardise”*

ZAMBIA QUALIFICATIONS AUTHORITY

Finsbury Park, Ground Floor, Kabwe Roundabout
P.O Box 51103, Lusaka, Zambia

+260 211 843 050 // +260 963 922 730
+260 956 037 185 // +260 972 559 301

info@zaqa.gov.zm
www.zaqa.gov.zm

<https://www.linkedin.com/company/zambia-qualifications-authority/>
<https://www.facebook.com/ZAQA2016/>

